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RESEARCH

UrologiQ: AI-based accurate detection, measurement and reporting of stones in CT-KUB scans

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Abstract
Kidney stone disease is becoming increasingly common worldwide, with its prevalence increasing annually across all age groups, races, and geographic regions. This sharp increase may be due to significant changes in dietary habits. Early and accurate detection of kidney stones is crucial for timely intervention and prevention of complications. This article discusses the role of artificial intelligence (AI) in detecting kidney stones and managing surgical treatments. Recent advances in AI techniques have introduced new tools that improve the diagnosis and analysis of medical images. AI can use CT-KUB image data to accurately detect the location of kidney stones and measure their size more efficiently than manual methods. AI-based detection methods ensure greater precision and consistency in stone identification and measurement. These improvements can help doctors plan treatments more effectively, resulting in a higher success rate for patients. Integrating AI into kidney stone detection and analysis significantly improves treatment planning and patient management, leading to better patient outcomes and overall quality of healthcare.

Keywords Nephrolithiasis · Percutaneous nephrolithotomy (PCNL) · Retrograde intrarenal surgery (RIRS) · Non-contrast computed tomography scan of the kidneys, ureter, and bladder (NCCT-KUB)

Introduction

Kidney stone disease, nephrolithiasis, and kidney stones are among the most significant health issues in India, affecting approximately 12% of the population [1]. Factors contributing to the development of kidney stones, also known as calculogenesis, include a lack of hydration, obesity, surgical procedures (such as surgery), high intake of salt or sugar, genetic predisposition in families, geographical location, climatic conditions, dietary habits, mineral content, and water consumption. These factors are associated with kidney diseases, diabetes, hypertension, and renal failure. Dealing with kidney stones poses a health challenge that requires a thorough understanding of how these stones form to prevent recurrence [2].

The signs of kidney stones vary according to their location in the kidney, ureter, or bladder. Common symptoms include blocked urine flow, dilatation of the calyx, also known as hydronephrosis, blood in the urine (hematuria), and episodes of renal colic affecting the urinary tract. Compared with women, men aged between 20 and 49 years tend to experience kidney stones [3].

Kidney stones can be categorized into four groups according to their mineral composition: cystine stones, uric acid stones, struvite or magnesium ammonium phosphate stones, and calcium stones (calcium phosphate and oxalate). Approximately 80% of kidney stones are calcium-based. Calcium oxalate (CaOx) accounts for half of all kidney stones, calcium phosphate (CaP) accounts for 5%, and a combination of both accounts for 45% [4].

Compared with other types of stones, calcium-based stones are more prevalent and are formed due to various factors. The struvite stones consisted of magnesium ammonium phosphate (MgNH4PO4.6H2O). Making up 10–15%

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of all kidney stones. These stones were formed by the combination of three cations (calcium, magnesium, and ammonium) and an anion (phosphate). Struvite stones are also referred to as infection stones because they are associated with tract infections. If left untreated, struvite stones can lead to health issues [4].

Uric acid stones account for approximately 3–10% of all stone types. Hyperuricosuria can result from the consumption of foods rich in purines such as meat. This shows that the excess uric acid in the urine is too high and that it also has a low volume and acidity (pH < 5.05), which increases the risk of kidney stone formation from uric acid [5].

Approximately 2% of all stones were cysteine stones. Cystinuria, a genetic disorder involving the transport of the amino acid cystine, causes excessive cystine in the urine and the formation of cystine stones. Treating these stones may require specialized treatment because they can obstruct the flow of urine and cause pain and complications [6].

The chemical composition of kidney stones can be divided into crystalline and nanocrystalline phases (organic and matrix stones, respectively). Matrix stones comprise macromolecules, such as proteins, lipids, carbohydrates, and glycosaminoglycans (GAGs). Matrix stones consist of proteins (64%), non-amino sugars (9.6%), glucosamine (5%), water (10%), and inorganic ash (10.4%) [7].

Preventive methods for treating kidney stones include increasing water intake, limiting foods that are high in oxalates, consuming calcium from food, maintaining moderate protein intake, avoiding high salt intake, and administering high doses of vitamin C.

Kidney stones are diagnosed when a patient experiences severe pain on the side, back, or below the ribs. The stones block the flow of urine in the kidneys, causing blood in the urine (hematuria) and urinary tract infections. The doctor recommended examinations and blood samples for urinalysis to assess the patient's signs. However, kidney stones can only be diagnosed via radiological imaging techniques through kidney, ureter, and bladder (KUB) scans via X-ray, CT-KUB, or ultrasound to determine the severity of kidney stones.

The stones often disappear without further treatment. However, if stones are > 3 mm in size, urinary tract infections can occur, and surgery or treatment for kidney stones is needed. The surgical treatment of kidney stones can be divided into four types according to the recommendations of the American Urological Association. Surgical treatment options include percutaneous nephrolithotomy (PCNL), ureteroscopy or retrograde intrarenal surgery (RIRS), shock wave lithotripsy, and open surgery.

Under general anesthesia, PCNL is used to remove large or irregularly shaped kidney stones. A flexible nephroscope is inserted into the posterior incision, and the kidney stone can be removed via a tube, shock waves, or suction device depending on fragmentation.

In RIRS, a ureteroscope is inserted through the urethra and bladder until it reaches the specific area where the kidney stone is located. This medical device is equipped with a laser fiber that effectively breaks the stone into smaller pieces. RIRS is primarily used to remove relatively small kidney stones, whereas stents are usually placed in the ureter to facilitate smooth urine flow.

Lithotripsy is a non-invasive surgical method that uses ultrasound or X-rays to locate stones. Large stones with difficulty passing through the urinary tract can be treated non-invasively via lithotripsy. used fluoroscopy to locate stones and focused shocks and ultrasound waves. Shock waves are used to break larger stones into smaller pieces, which then pass through the urinary system.

In addition, doctors recommend open surgery only if kidney stones cannot be removed via other methods. It is important to locate the stone and make an incision to minimize bleeding and prevent serious damage to major organs, blood vessels, and kidneys.

Materials and methods

A retrospective study was conducted to investigate and validate the AI-based UrologiQ model for the automatic detection and quantification of kidney stones in CT-KUB scans in patients with urinary stone disease [8, 9] and to investigate the factors influencing the need for emergency urological procedures and the prediction of the necessary interventions influencing treatments [10–12].

CT KUB (Kidneys, Ureters, and Bladder) scans were performed using CT scanners with an image resolution of 512 × 512 pixels per slice. The scans had a slice thickness of ≤ 2.5 mm and a slice spacing of ≤ 2 mm, covering the abdominal area using a soft convolution kernel at 120 kV and 180 mAs.

From June 2023 to January 2024, 600 NCCT-KUB scans were collected from eleven different diagnostic centres by transferring data from the centre's local PACS system to a cloud-based UrologiQ server. The dataset contained 3D NCCT KUB scans of patients over 18 years of age who were diagnosed with normal KUB systems without stones ($n = 100$), and stones were detected in their scans ($n = 500$). The distribution of clinical patients from the centres, classified as normal or abnormal by both radiologists and the AI model are shown in Supplementary Tables 1a and 1b.

Kidney stones with a maximum dimension of ≥ 2 mm were included in the analysis. Patients who underwent double-J stent placement or post-PCNL procedures were also included. The exclusion criteria for the dataset were

as follows: (1) NCCT-KUB scans of pediatric patients; (2) congenital renal anomalies (e.g., duplex ureters, horseshoe kidneys, and ectopic kidneys); (3) renal anomalies (e.g., polycystic kidneys, tumors, and masses); (4) abdominal abnormalities that may obscure the ureteral tract (e.g., ascites or pneumoperitoneum); and (5) CT scan artifacts (e.g., streaking and motion artifacts).

Ascites were used as an exclusion criterion because the software is designed for simple CT images where tracking the ureter can be challenging. Additionally, intestinal or arterial calcifications could be misidentified as stones. In clinical practice, such cases typically require a contrast study tracing the ureter in the delayed phase. However, because the software does not analyze contrast-enhanced scans, hardening of the arteries may be mistakenly identified as calculi, leading to false positive results.

When perforation is used as an exclusion criterion, this specifically means intestinal perforation, as tracing the ureter on simple CT scans can be difficult in such cases. Although rare, renal perforations, commonly seen on postoperative scans, can also result in false-positive results. Due to the rarity of these cases, the AI model was not trained on them and they were therefore excluded from the analysis.

In this study, data were collected in DICOM format, which contained patients' personal information. To protect patient privacy, an anonymization code was used to remove personal information from the DICOM tags. After anonymization, the files were converted to the NIFTI format. Additionally, each case was assigned a unique identification code to ensure data confidentiality and traceability. The data were then processed via UrologiQ 2.0, and the results were recorded.

Data analysis

In this retrospective study on the automatic detection and quantification of kidney stones, we included only CT-KUB scans. CT images that did not originate in the KUB area were excluded. Individuals with congenital diseases, horseshoe kidney, or duplex ureter, were excluded from the study. Images showing ascites or perforations in the KUB area were excluded. The scans included in this study had an XY dimension range of 512×512 pixels and a slice thickness of ≤ 2.5 mm. However, there were differences in the XY distance from $0.6 \times 0.6 \times 0.8$.

The UrologiQ artificial intelligence (AI) algorithm architecture (Fig. 1) consists of four steps: preprocessing, inference, postprocessing, and reporting. In the first step, segmentation masks for abdominal organs (kidneys, ureters and bladder) are generated using a 3D UNET architecture based on convolutional neural networks (CNNs). After obtaining the organ masks, an algorithm filters out stone masks with Hounsfield Units (HU) above a threshold of 140 within the segmented regions. Each scan undergoes preprocessing to improve generalization, including resampling to a standard resolution and normalization. We also integrate scans with different slice thicknesses into the training data to further improve the robustness of the model. To begin the analysis, we removed cases in which the kidneys were < 10 cm3 in size or volume.

The images were smoothed according to the noise level. The input CT-KUB scans first go through a Gaussian noise filter algorithm during preprocessing. It uses a noise threshold below which a smoothing factor of 0.5 is applied to the images. If the noise exceeded this threshold, a smoothing factor was triggered. In the inference phase, we use a 3D CNN-based U-NET model [13] to accurately segment abdominal organs such as the kidney, ureter, and bladder.

After receiving the segmentation results, a limit of 140 HU (house field units) is maintained in the segmentation masks, and the stone masks are filtered accordingly. A pixel count-based rule is applied when the noise threshold is within a certain range. According to this rule, a stone mask was created only if the identified stone exceeded a certain threshold. However, stones that do not meet this requirement are not considered to be concrete.

Aortic calcifications were then filtered by overlapping them with aortic masks obtained from another CNN-based 3D UNET model. This process was performed in the postprocessing phase after each stone mask was obtained and positioned accordingly at the appropriate location.

The results of the processed algorithm are summarized in an automated report generated during the report-generation phase. The results for each case were entered into an NIFTI file via the UrologiQ AI algorithm.

The 600 NCCT-KUB scan reports (validation dataset) collected from the radiology centers were validated by radiologists, and the results were entered for each case individually. The sensitivity, specificity, and accuracy of the AI algorithm in identifying, locating, and measuring kidney stones were determined by combining these data with the AI results.

Results

Ground truth was determined by two radiologists reviewing the CT scan DICOM data and associated reports and reaching consensus on the assessment. To evaluate the agreement between radiologist assessments and AI predictions, we analyzed cases on the basis of their classification as normal or abnormal (with at least one stone present).

Specifically, we examined two scenarios:

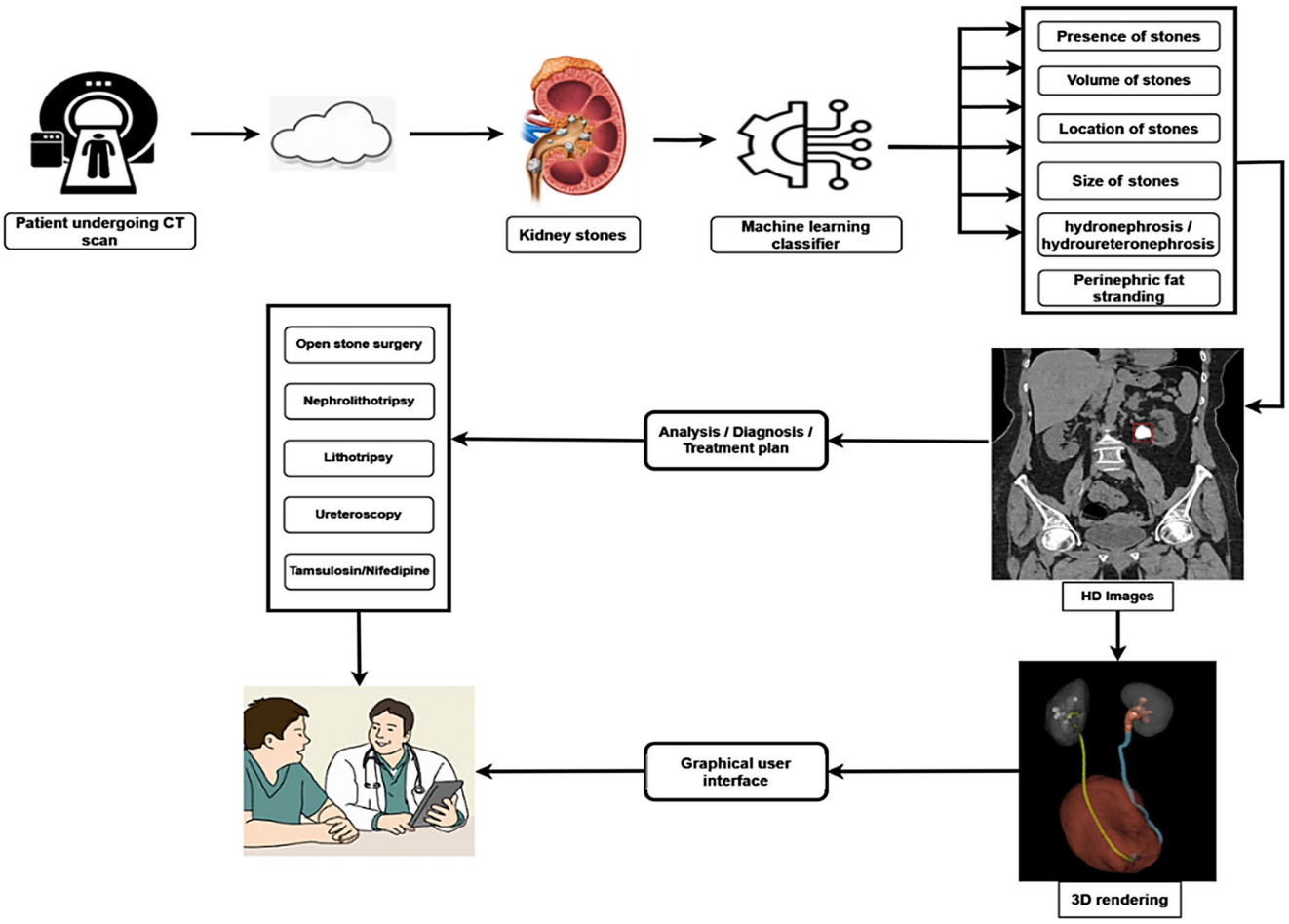

Fig. 1 Architecture of the UrologiQ algorithm for the detection of kidney stones in CT-KUB scans

Table 1 Examination of cases with kidney stone detection and comparison with radiologist and AI assessments

		Normal radiologist	Abnormal radiologist	Total
Normal AI	**Count**	104	3	107
	Percentage	97.19%	3%	100%
Abnormal AI	**Count**	3	489	492
	Percentage	0.60%	99.39%	100%
Total	**Count**	107	492	599
	Percentage	18%	81.13%	100%

1. If the radiologist classified a case as normal without stones, did the AI also predict it as normal?
2. Did the AI have the same abnormal classification when the radiologist identified a case as abnormal, indicating the presence of at least one stone?

The analysis represents the number of cases in which both the radiologist and AI agreed on the normal/abnormal classification, highlighting the accuracy and reliability of AI in replicating radiologists' assessments, as reported in Table 1; Fig. 2.

Understanding the performance metrics of AI compared with radiologist outcomes is critical for assessing its clinical utility and guiding its integration into healthcare systems. This study aimed to comprehensively evaluate the sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and accuracy of AI compared with those of radiology in medical diagnosis. AI algorithms and radiological assessments were independently applied to the dataset. Confusion matrices were created to calculate the sensitivity, specificity, PPV, NPV, and accuracy of both the AI and radiology results.

Sensitivity, also known as the true positive rate, measures the proportion of actual positive cases correctly identified by a diagnostic test or diagnostic model. In medical research, sensitivity refers to the ability of a test to correctly identify individuals with the disease being examined. It is calculated as the ratio of true positives to the sum of true positives and false negatives: A high sensitivity value indicates that the test rarely misses positive cases, making it valuable in ruling out the disease.

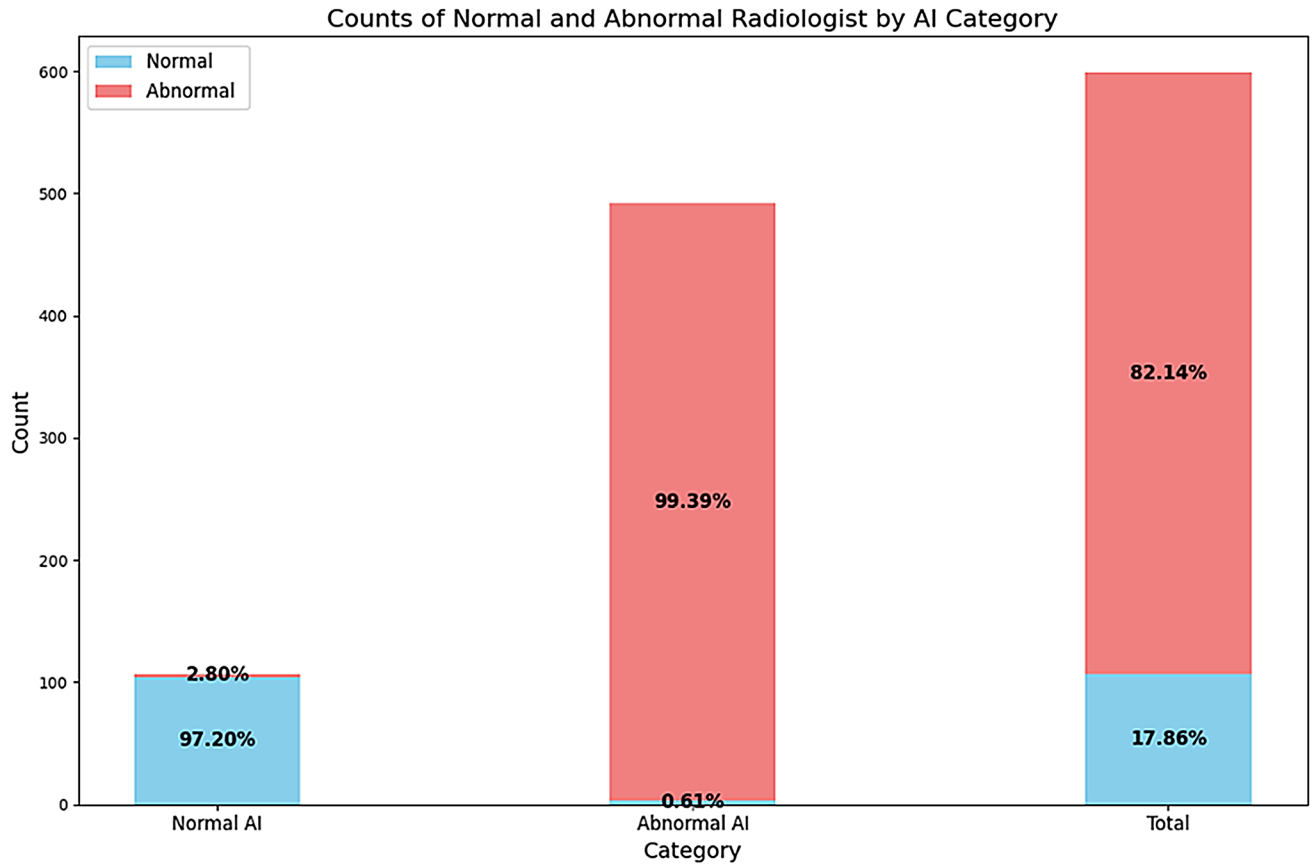

Fig. 2 Radiologist and AI assessments for kidney stone detection

The **specificity or true negative rate** assesses how well a diagnostic test or model correctly identifies cases that do not have the condition being tested. This demonstrated the accuracy of recognizing individuals without this condition. Specificity was determined by comparing the number of negatives with the total number of negatives and false positives. A high specificity value indicates that the test seldom misidentifies cases, making it useful for confirming the absence of a disease.

The **positive predictive value (PPV)**, known as precision, indicates the percentage of results among all positive outcomes from a diagnostic test or model. The PPV indicates the likelihood of a result indicating the presence of disease. It is computed by dividing the positives by the sum of the positives and the false positives. A high PPV suggests a positive result, which is beneficial for confirming the presence of disease.

The **negative predictive value (NPV)** assesses the proportion of outcomes among all negative results produced by a diagnostic test or model. The NPV signifies the likelihood that a negative test result indicates the absence of disease. It is computed as the ratio of negatives to the combined total number of false negatives. A high NPV suggests a negative result, making it valuable to exclude the presence of the disease.

Accuracy quantifies the correctness of a test or model and is determined by dividing the sum of true positives and true negatives by the total number of cases. Although accuracy offers an evaluation of performance, it may not be adequate in scenarios with imbalanced datasets or varying costs associated with positives/negatives. Hence, alongside accuracy, sensitivity, and specificity, the positive predictive value (PPV) and NPV are commonly delineated to facilitate nuanced performance evaluation.

The formulas for calculating AI performance metrics are as follows:

$$Positive\ Predictive\ Value\ (PPV) = \frac{TP}{TP + FP}$$

$$Negative\ Predictive\ Value\ (NPV) = \frac{TN}{TN + FN}$$

$$Sensitivity\ (Recall) = \frac{TP}{TP + FN}$$

Table 2 Case-level performance metrics

Performance metrics of Artificial Intelligence	Percentages
Sensitivity (Recall)	99.39%
Specificity	97.10%
Positive Predictive Value	99.39%
Negative Predictive Value	97.10%
Accuracy	99.0%
F1 Score	99.39%

$$Specificity = \frac{TN}{TN + FP}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

True positive (TP)

The AI detects the presence of kidney stones, and the radiologist confirms the presence of kidney stones.

False positive (FP)

AI indicates the presence of kidney stones. A radiologist reported no kidney stones.

Table 3 Stone-level performance metrics

Performance metrics of Artificial Intelligence	Percentages
Sensitivity (Recall)	98.96%
Positive Predictive Value (Precision)	96.48%

True negative (TN)

The AI identifies no presence of kidney stones, which aligns with the radiologists' findings of no kidney stones.

False negative (FN)

The AI failed to detect any kidney stones, while the radiologist diagnosed the presence of kidney stones.

In the assessment, artificial intelligence achieved true negative (TN) = 104, false positive (FP) = 3, false negative (FN) = 3, and true positive (TP) = 489 compared with the radiologists' findings for all cases. The details of the analyzed samples are presented in Table 2; Fig. 3.

The artificial intelligence scores were true positive (TP) = 1426, false positive (FP) = 52, and false negative (FN) = 15 compared with the radiologist's result for the total stones. The analyzed samples are listed in Table 3.

We analyzed the distribution of the key calculus locations identified by both radiologists and artificial intelligence (AI) systems. The main objective of this study was to compare the accuracy and consistency of AI for detecting kidney stones at different anatomical locations. The dataset included various locations, such as the renal pelvis, ureters, and bladder. Radiological assessment is the gold standard for evaluating AI predictions.

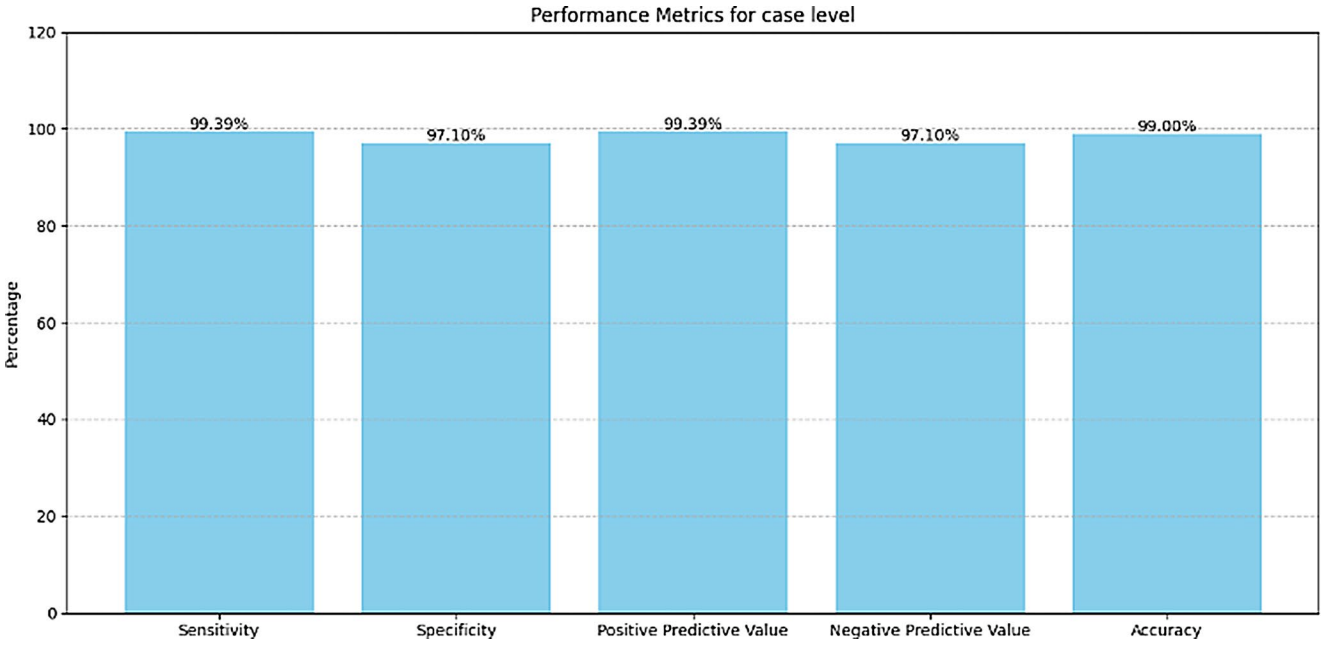

Fig. 3 Case-level performance metrics

Our findings demonstrated that the AI exhibited a high degree of concordance with radiologists in identifying calculus locations, particularly in the renal pelvis and bladder. However, some discrepancies were noted in the ureter, where the AI occasionally missed smaller stones or misclassified their exact location. These results underscore the potential of AI to assist in clinical settings, although they also highlight areas for improvement to increase the precision of AI-based diagnostic tools in accurately identifying and localizing kidney stones, the distribution of which is shown in Table 4.

This study examined the correlation between longitudinal (LR), anteroposterior (AP), and superoinferior (SI) measurements of kidney stones determined via artificial intelligence (AI) and those measured by radiologists.

This study aimed to evaluate the precision of AI in replicating the measurements determined by radiologists, which serve as a clinical standard. Statistical analysis revealed a strong positive correlation between the AI and radiologist measurements in all three dimensions (LR, AP, and SI), suggesting that the AI can reliably approximate the size and orientation of kidney stones.

However, minor deviations were observed, particularly for irregularly shaped stones, where the AI measurements occasionally differed slightly from those of radiologists. Overall, these results support the potential utility of AI in clinical practice by providing accurate and consistent measurements that closely match those of experienced radiologists (Table 5).

Scatterplots were used to visualize the relationships between the AI-derived measurements and radiologist measurements for the longitudinal (LR), anteroposterior (AP), and superoinferior (SI) dimensions of the kidney stones. These scatterplots illustrate a strong linear correlation, with data points clustered closely around the line of best fit, indicating high agreement between the AI and radiologist measurements.

The slopes of the regression lines were close to unity, further confirming the accuracy of AI in mirroring radiologists' assessments. The outliers were minimal, suggesting that the AI predictions were generally reliable, although slight variations were observed for atypically shaped stones. Overall, the scatterplots highlight the effectiveness of AI in providing measurements that are consistent with radiologists' assessments, as shown in Fig. 4.

Boxplots were created to compare the AI and radiologist measurements of kidney stone size in the longitudinal (LR), anteroposterior (AP), and superoinferior (SI) dimensions. These boxplots demonstrate a high level of agreement between the AI and radiologist measurements, with the medians and interquartile ranges (IQRs) closely matching across all dimensions, as reported in Table 6.

Table 4 Calculus location distributions obtained by radiologists and artificial intelligence

		Radiologist		AI	
Side	Location	Count	%	Count	%
Bladder	Bladder	10	1	10	1
Left Kidney	PUJ	13	1	13	1
Left Kidney	Lower Calyx	204	15	214	15
Left Kidney	Middle Calyx	207	15	187	13
Left Kidney	Renal Pelvis	20	1	19	1
Left Kidney	Upper Calyx	125	9	136	10
Left Ureter	UVJ	26	2	26	2
Left Ureter(lower)	Lower Ureter	44	3	45	3
Left Ureter(middle)	Middle Ureter	17	1	17	1
Left Ureter(upper)	Upper Ureter	45	3	46	3
Right Kidney	PUJ	19	1	19	1
Right Kidney	Lower Calyx	155	11	165	12
Right Kidney	Middle Calyx	188	14	184	13
Right Kidney	Renal Pelvis	18	1	16	1
Right Kidney	Upper Calyx	123	9	117	8
Right Ureter	UVJ	35	3	34	2
Right Ureter(lower)	Lower Ureter	51	4	51	4
Right Ureter(middle)	Middle Ureter	25	2	24	2
Right Ureter(upper)	Upper Ureter	66	5	68	5

Table 5 AI correlations between radiologist measurements and LR, AP, and SI measurements

Artificial intelligence	Correlation (*r*)	*P* value*
LR	0.989	< 0.001
AP	0.985	< 0.001
SI	0.986	< 0.001

* Spearman's rank correlation coefficient

The spread of the data, as indicated by the whiskers, showed that the AI measurements were consistent with those of the radiologists and that there were few outliers. This close agreement in the boxplots highlights the ability of AI to accurately measure kidney stone size and highlights its potential as a reliable tool for clinical diagnostics (Fig. 5).

The effectiveness of the AI in predicting the presence, location, and size of kidney stones was evaluated via receiver operating characteristic (ROC) curves and compared with assessments by radiologists.The ROC curves revealed high **area under the curve** (AUC) values for all dimensions—longitudinal (LR), anteroposterior (AP), and superoinferior (SI)—indicating that the AI model has exceptional discrimination ability, as shown in Fig. 6.

AI is very accurate in determining the location and size of stones, as well as differentiating between cases with and without stones, on the basis of AUC values that are very close to 1.0. These results highlight the robustness of the AI system in adapting it to the diagnostic precision of radiologists, thereby confirming its effectiveness in clinical applications.

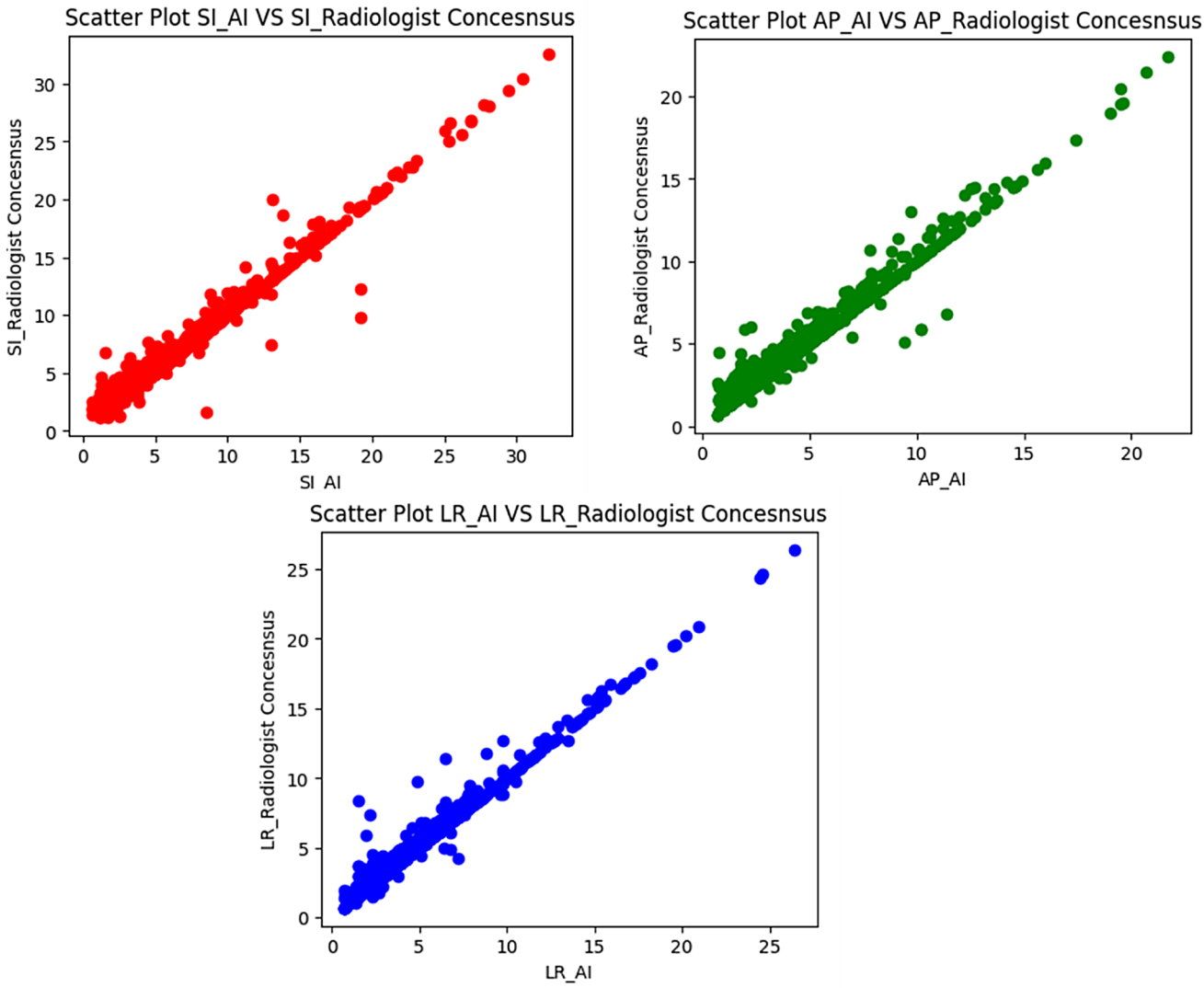

Fig. 4 Scatterplots comparing AI-derived measurements and radiologists' consensus on kidney stone size

Table 6 Kidney calculus size measurements via artificial intelligence and radiologists

	Mean	Std deviation	Q1	Median	Q3	Min	Max
Artificial Intelligence							
SI	5.9	4.53	2.92	4.3	7.5	0.6	32.2
LR	4.29	2.9	2.3	3.5	5.4	0.7	21.7
AP	4.26	2.83	2.3	3.5	5.4	0.7	20.7
Radiologist							
SI	6.15	4.51	3.1	4.6	7.6	1.2	32.6
LR	4.38	3.2	2.3	3.4	5.5	0.7	26.4
AP	4.45	2.92	2.4	3.65	5.6	0.7	22.43

Table 7 shows the area under the curve (AUC) results when the AI is compared. Radiologists assess calculus stone presence, location, and size in terms of the LR, AP, and SI dimensions. The AUC value was 0.980, indicating the accuracy of the AI system compared with that of human radiologists. With an error of 0.008 and a p-value < 0.001, the high AUC was statistically significant and not merely by chance. The 95% confidence interval for the AUC fell between 0.966 and 0.997, highlighting the consistency and precision of the AI measurements in detecting calculus stones. In summary, this table underscores the performance of AI algorithms in identifying and measuring calculus stone rivals or even surpasses the capabilities of human radiologists.

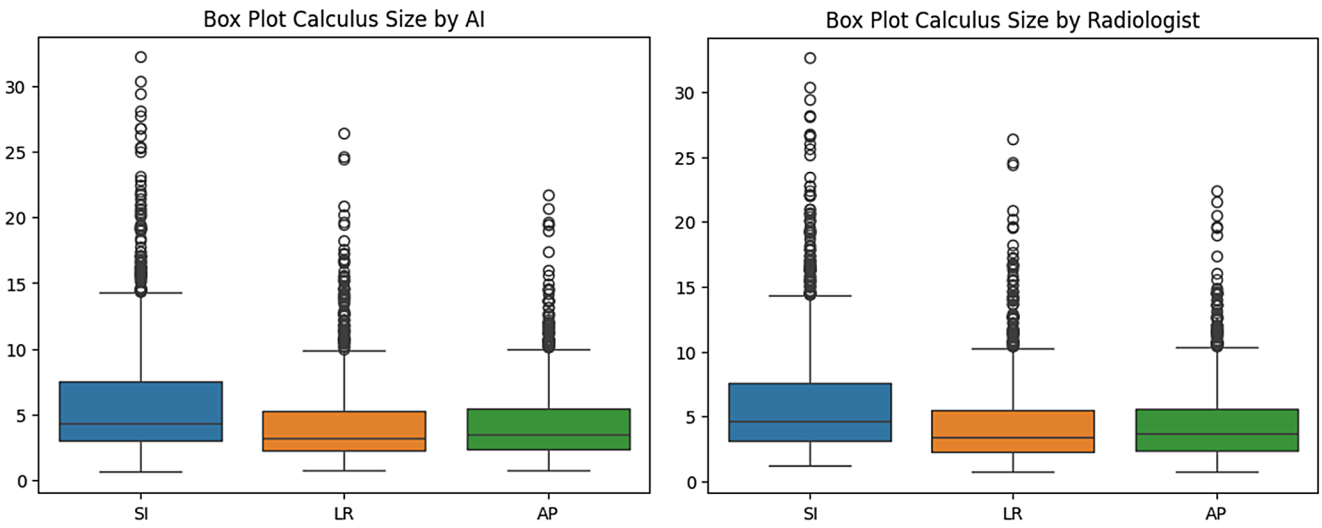

Fig. 5 Comparison of AI and radiologist measurements of kidney stone size in different dimensions

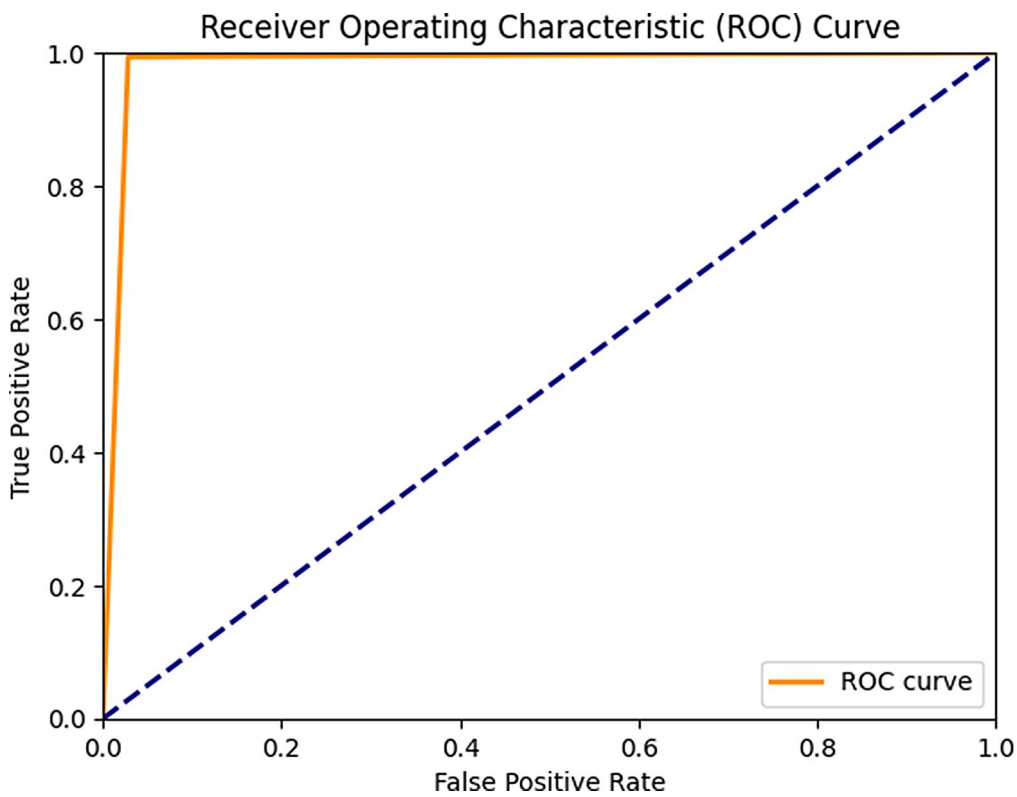

Fig. 6 Case-level performance metrics

Table 7 Area under the curve (AUC) for AI vs. radiologist measurements of calculus stone presence, location, and size (LR, AP, and SI dimensions)

Area	Std. Error	*P* value	95% Confidence interval (AUC)	
			Lower bound	Upper bound
0.980	0.008	< 0.001	0.966	0.997

Conclusion

Nephrolithiasis is a common problem with an increasing prevalence. The use of artificial intelligence in CT-KUB scans to automatically detect and report kidney stones has led to significant progress in terms of faster reporting and diagnostic accuracy. The algorithm can effectively identify and classify different types of stones with high precision,

helping urologists and radiologists achieve faster and more reliable results for patients. Our study highlights the potential of AI algorithms to consistently outperform traditional methods, particularly in complex cases in which detecting smaller or less conspicuous stones is critical.

The results of our experiments underscore the importance of training AI models with extensive and diverse datasets to improve their robustness and generalizability across different patient populations and imaging modalities. Therefore, future developments in AI-assisted stone detection are promising. One significant area of progress is the refinement of the algorithm to further reduce false positives and negatives, thereby increasing diagnostic confidence.

Additionally, advances in the interpretability of AI are critical. The development of models that provide a clear and understandable rationale for diagnostic decisions will increase trust and acceptance among physicians. Collaboration among AI researchers, clinicians, and regulators is critical to ensure that these systems are safe, effective, and aligned with clinical needs.

In summary, the use of AI for stone detection is a transformative step in medical diagnostics. Through continued research, technological innovation, and interdisciplinary collaboration, AI has the potential to revolutionize the field and lead to better patient care and outcomes.

AI is still in development to predict the type of surgery based solely on CT scans and assess the likelihood of stone passage taking into account factors such as stone size, location and ureter dimensions. In addition, it aims to predict the success rate of ureteroscopy (URS) and help determine the optimal puncture site for percutaneous nephrolithotomy (PCNL) based on calyx and kidney anatomy. Future advances should focus on improving algorithmic accuracy by incorporating a larger and more diverse sample size, optimization for low-dose CT images, stent stone detection, improving ureteral and bladder stone classification, accurate stone composition identification, and refinement Prioritize predicting surgical decisions. It is also important to ensure that these innovations are accessible and beneficial to diverse patient groups worldwide.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethical approval The principles of the Declaration of Helsinki were followed in lieu of a formal ethics committee review.

Competing interests The authors declare no competing interests.

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